Hidden Pressures on NSW Breeding Birds

Five further questions from the Atlas and DES records

Client briefing · CS-13 Combined Breeding Analysis team · 18 October 2026

Data: NSW Bird Atlas (1977–2020) and DES backlog (to 2025)

THE FIVE QUESTIONS

Q1 **Do small woodland birds breed less where Noisy Miners are present?**

Where miners are present, small woodland birds are recorded 7% less often and their odds of breeding are 10% lower; large birds are unaffected.

Q2 **Which introduced birds are spreading across NSW, and which are retreating?**

The Common Myna went from 10% to 24% of squares and is moving inland; sparrows, starlings and goldfinches retreated.

Q3 **Which native birds are winning and which are losing ground, on a fair like-for-like comparison?**

9 of the 12 native birds that lost most ground eat insects, such as Black-faced Woodswallow and Hooded Robin; none of the 12 biggest winners do.

Q4 **Are migratory birds that leave NSW each winter declining faster than birds that stay?**

79% of summer migrants did worse than typical, against 44% of non-insect-eating residents; insect-eaters that stay all year also slipped.

Q5 **How few places would protect breeding for almost every species, and are they still being watched?**

32 grid squares hold recent breeding for 90% of native species; 10 of them had no survey in 2021–2025.

HOW TO READ THIS REPORT

This report is a companion to 'Understanding Bird Breeding in NSW'. Each page asks one research question, answers it with a chart, and explains what it means for protecting species. Because survey lists got shorter over time, comparisons are made like-for-like and against the typical species, not on raw totals.

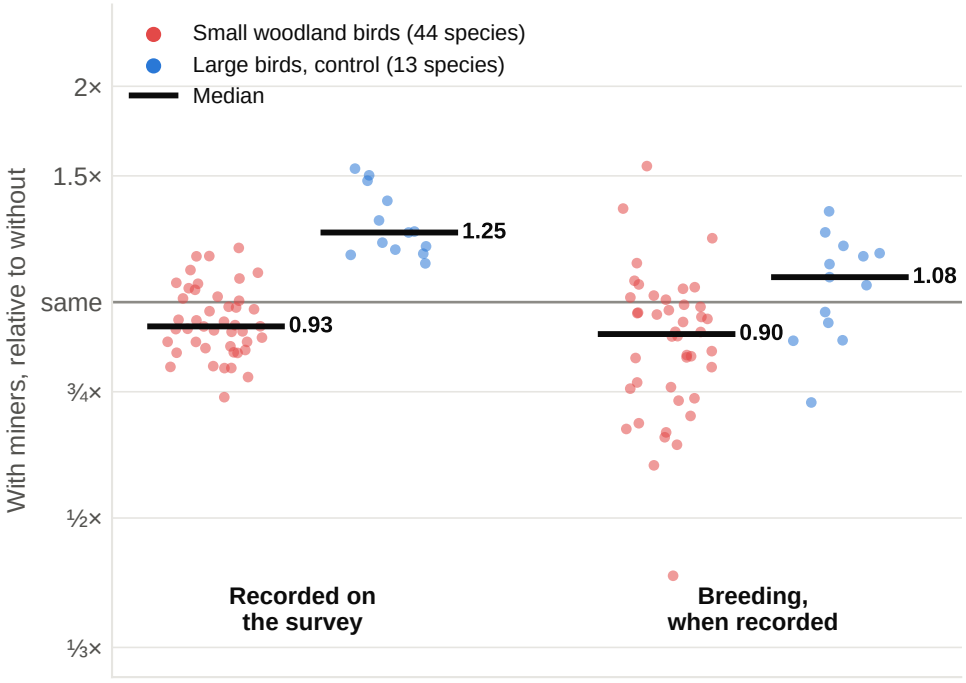
RESEARCH QUESTION 1

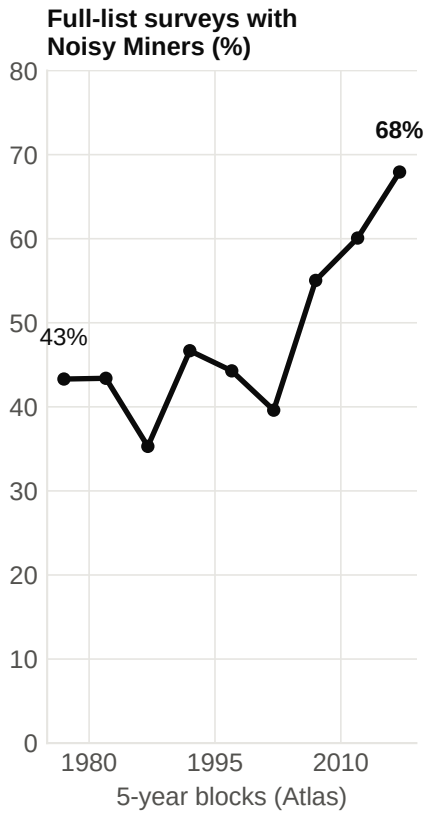
Do small woodland birds breed less where Noisy Miners are present?

Why this matters: Noisy Miners live in groups and aggressively drive other birds from their territory; this is a listed Key Threatening Process in NSW. The records can show whether small birds breed less where miners are.

With Noisy Miners present, small birds' odds of breeding are 10% lower

Surveys with vs without miners in the same square, period and list length ('same' = no difference)





ANSWER

- On otherwise-alike surveys, small woodland birds were recorded 7% less often when miners were present, while large birds were recorded 25% more often.
- When small birds were recorded alongside miners, the odds that they were breeding were about 10% lower. 33 of 44 small species showed lower breeding; large birds showed no such drop (median 1.08).
- Among species with at least 100 breeding records, the biggest breeding gaps were for Flame Robin, Eastern Yellow Robin, White-eared Honeyeater and Crested Shrike-tit.
- Miners became more common: they were on 43% of full-list surveys in 1975–1979 and 68% in 2015–2019.

What it means: The pattern matches the known impact of Noisy Miners on small birds. Restoring a dense shrub layer, and miner control in high-value woodland remnants, could lift breeding of many small species at once.

Caution: Miners prefer open, fragmented woodland, and that habitat may itself suit small birds less. The matching removes place and time differences but cannot fully separate the miners from the habitat they choose.

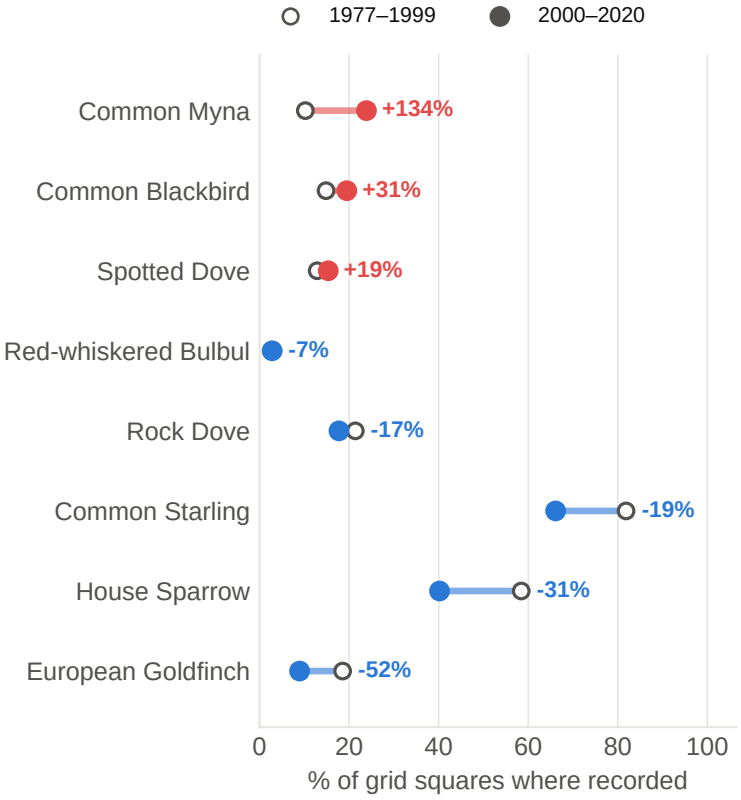
RESEARCH QUESTION 2

Which introduced birds are spreading across NSW, and which are retreating?

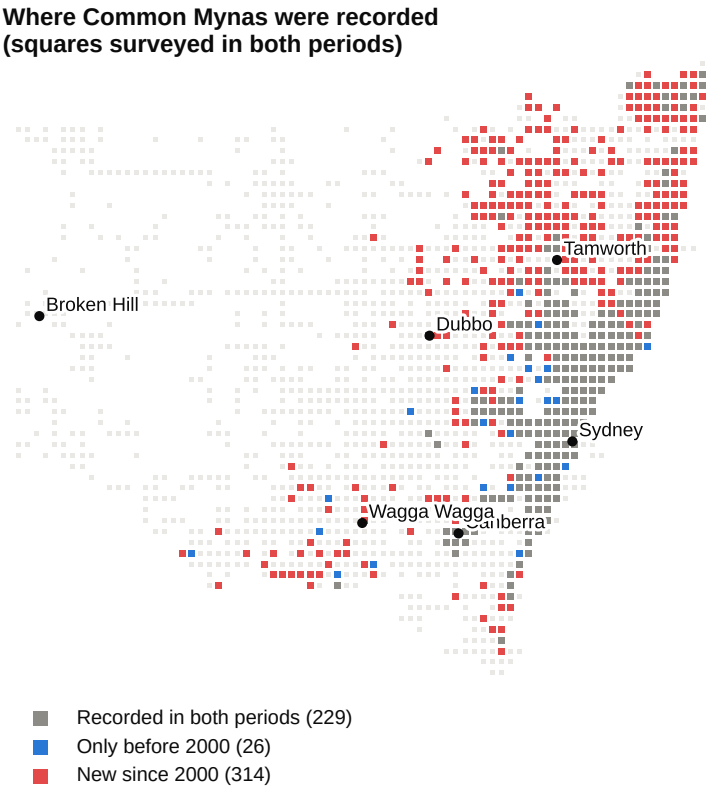
Why this matters: Introduced birds compete with natives for food and nest sites. The Common Myna in particular takes over tree hollows that native parrots, kingfishers and gliders need. Knowing what is spreading shows where control could stop new invasions.

The Common Myna more than doubled its range, from 10% to 24% of squares

Share of grid squares where each introduced bird was recorded (fair like-for-like comparison)



Typical native species changed by -13%



ANSWER

- On a fair like-for-like comparison, the Common Myna went from 10% to 24% of grid squares (+134%), while the typical native bird changed by -13%.
- The Myna appeared in 314 new squares and disappeared from only 26. West of the Great Dividing Range (west of 150°E) its squares went from 36 to 144, so it is moving inland along towns and farmland.
- Common Blackbird and Spotted Dove gained more modestly (+31%, +19%). European Goldfinch, House Sparrow, Common Starling, Rock Dove retreated faster than the typical native bird.
- Breeding records of House Sparrows and Common Starlings fell from 319 and 504 squares to 109 and 169, while the Myna bred in 64 then 77 squares.

What it means: The Common Myna is the introduced bird to act on. Trapping and nest-box programmes are most useful at the inland edge of its range, before it settles in new towns and competes for hollows.

Caution: Introduced birds live mostly near people, and volunteer surveys visit towns unevenly; the like-for-like comparison reduces but does not remove this effect.

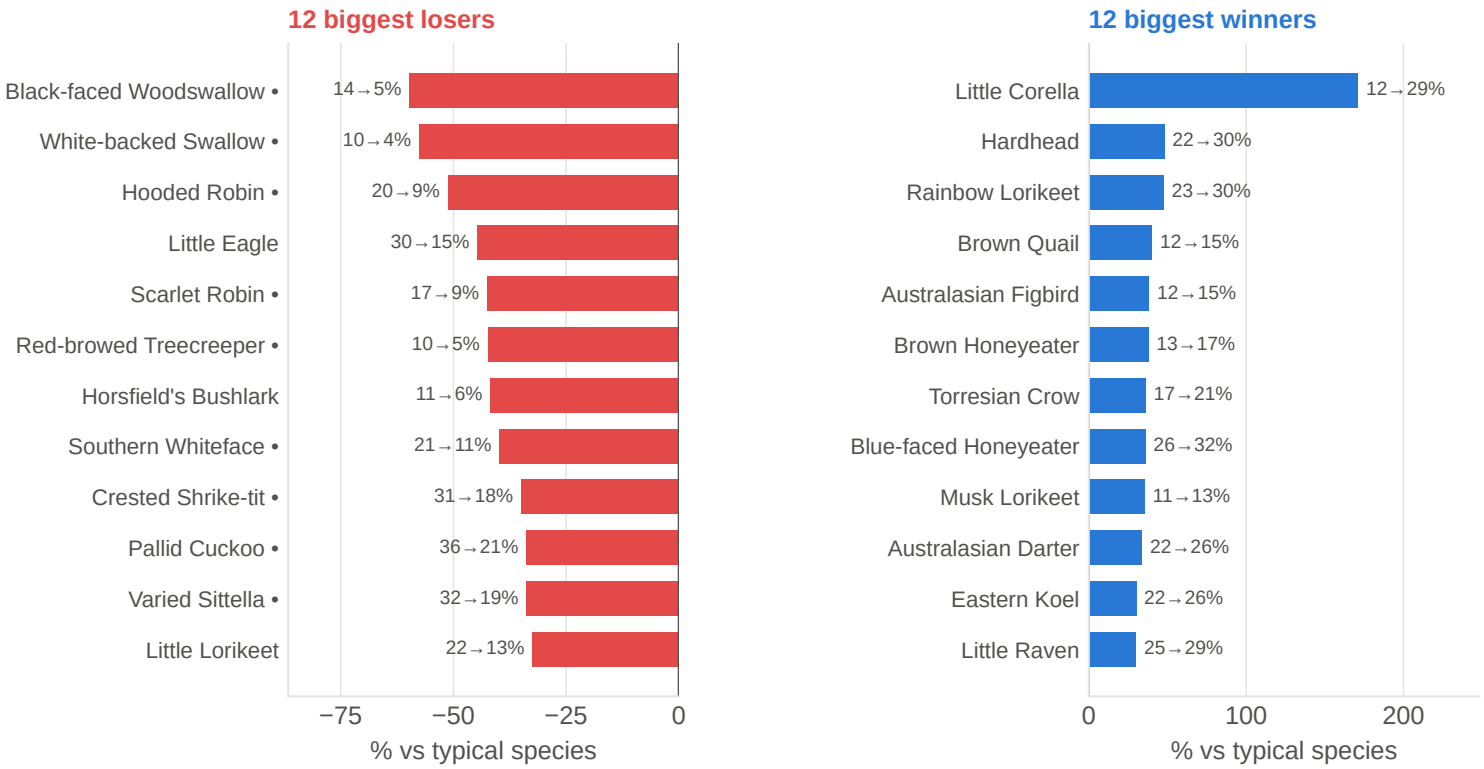
RESEARCH QUESTION 3

Which native birds are winning and which are losing ground, on a fair like-for-like comparison?

Why this matters: The kinds of birds that win and lose say more about what is changing in the landscape than any single species. Losers that share a way of life probably share a pressure that can be managed together.

9 of the 12 biggest losers eat insects, but none of the 12 biggest winners do

Change in share of grid squares where recorded, Sep–Mar, vs the typical species (fair comparison)



• = mainly eats insects. Labels show the share of squares where recorded, then → now.

ANSWER

- Of 194 widespread native species, 24 lost at least 25% more ground than the typical species and 14 gained at least 25% more.
- The biggest losers are Black-faced Woodswallow, White-backed Swallow, Hooded Robin, Little Eagle, Scarlet Robin, Red-browed Treecreeper. 9 of the 12 biggest losers mainly eat insects, most of them birds of open woodland and grassland.
- The biggest winners are Little Corella, Hardhead, Rainbow Lorikeet, Brown Quail, Australasian Figbird, Brown Honeyeater. None of the 12 biggest winners mainly eat insects. They are a mix of 3 parrots and lorikeets, water birds, and fruit- and nectar-eaters common in gardens and towns.
- Together this points to insect-eating woodland birds giving way to more adaptable species.

What it means: Protecting and restoring open woodland with native grasses, fallen timber and insect-rich ground layers would help the largest group of losers at once. Q4 looks at whether migrants are a special case.

Caution: 'Winning' and 'losing' here mean recorded in more or fewer places, not the number of birds. Shorter recent lists lower all species, which is why results are compared with the typical species.

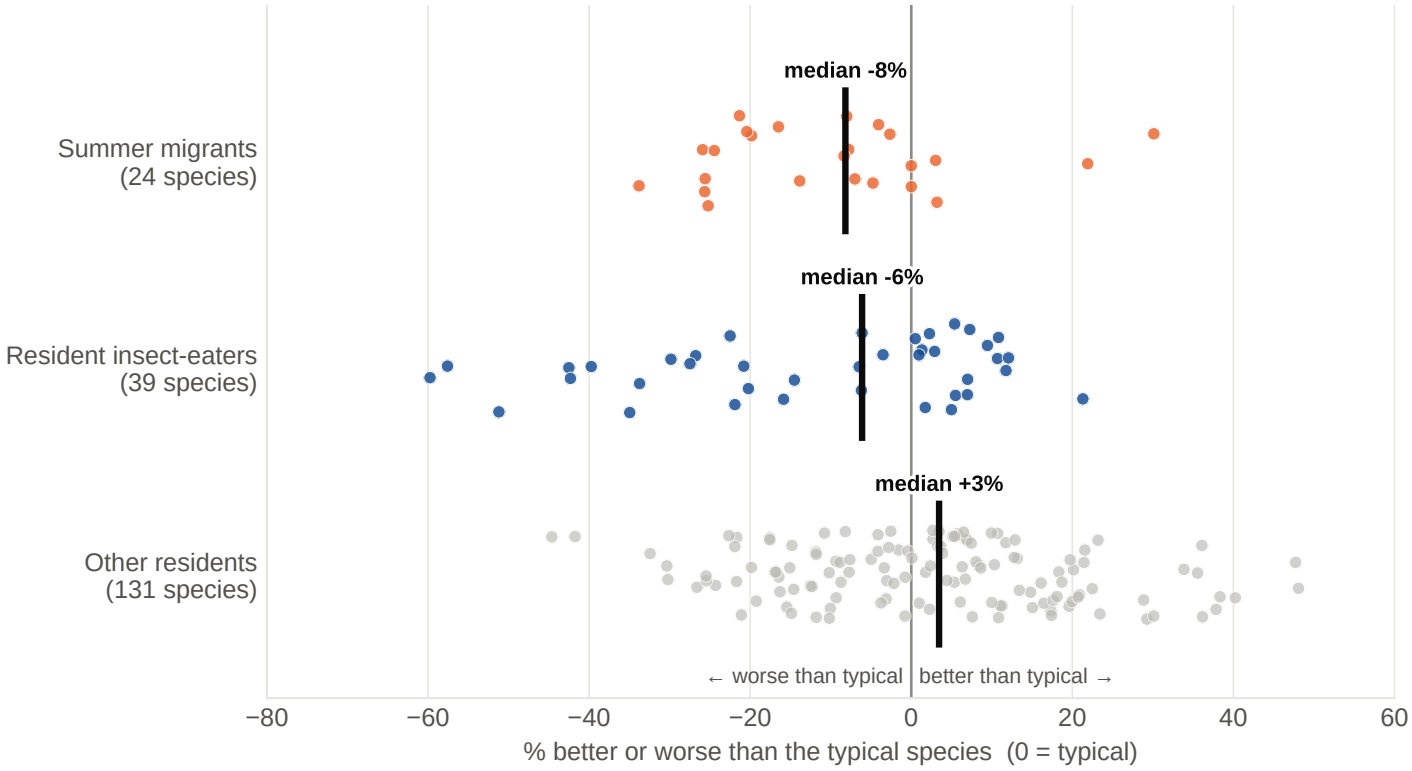
RESEARCH QUESTION 4

Are migratory birds that leave NSW each winter declining faster than birds that stay?

Why this matters: If birds that winter in northern Australia or New Guinea are declining faster than those that stay, part of the cause lies outside NSW — on migration routes or wintering grounds — and needs action beyond NSW.

79% of summer migrants did worse than the typical species

Change in share of grid squares where recorded, Sep–Mar, relative to the typical species (each dot = species)



ANSWER

- Summer migrants lost more ground than other birds: median -8% against the typical species, versus +3% for other residents (difference unlikely to be chance, p = 0.001).
- 79% of the 24 migrant species did worse than typical, compared with 44% of other residents.
- Resident insect-eaters also slipped (median -6%), and the gap between them and migrants is not clear (p = 0.7). The decline is shared by insect-eaters in general, with migrants the worst affected.
- The steepest migrant losses were Pallid Cuckoo, Brown Songlark, White-browed Woodswallow, Tree Martin.

What it means: Insect-eating birds, and summer migrants most of all, are the groups to watch. Possible pressures include fewer insects in farmland and woodland in NSW, and, for migrants, conditions on their routes and wintering grounds. The migrant results are worth sharing with national migratory-bird programmes.

Caution: The migrant list is based on general knowledge of each species' movements; some species are only partly migratory. The comparison cannot say where along the journey the pressure lies.

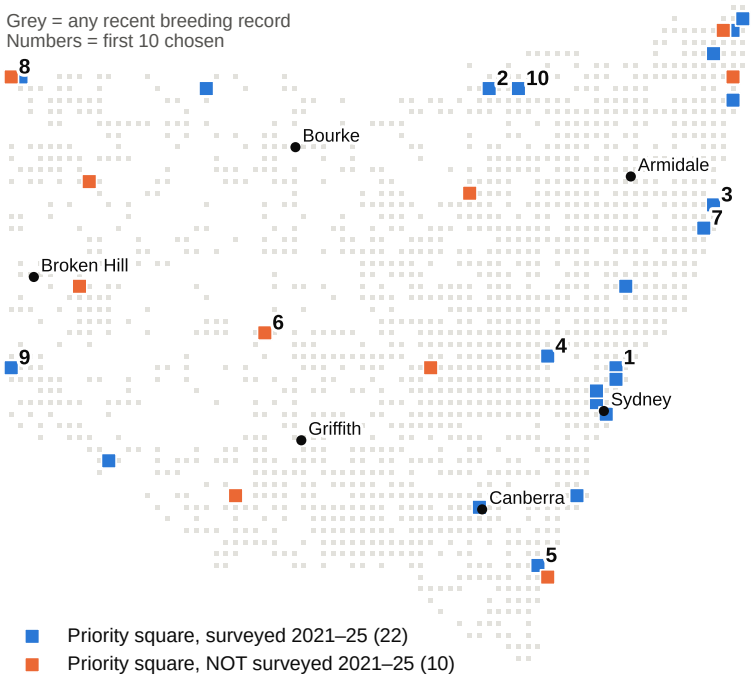
RESEARCH QUESTION 5

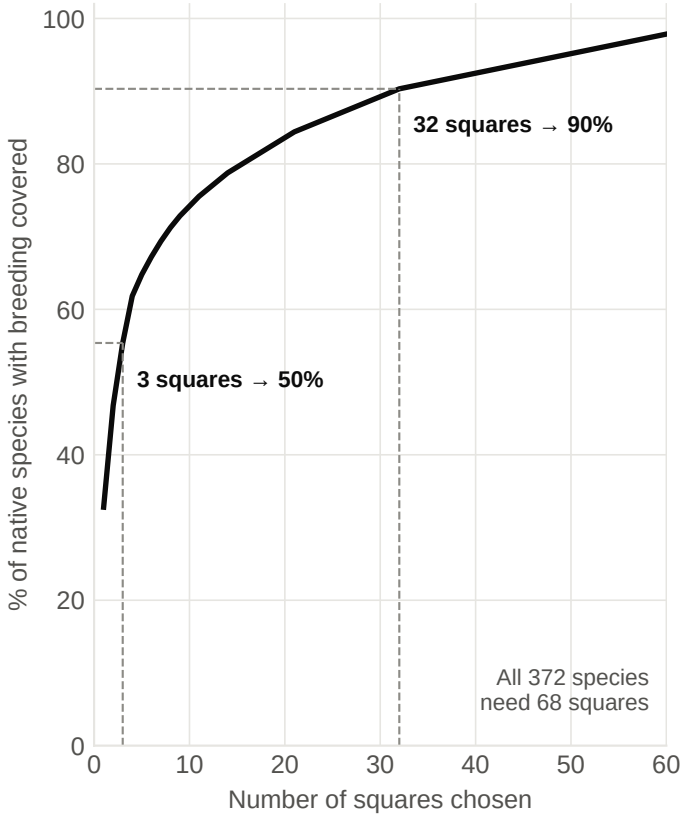
How few places would protect breeding for almost every species, and are they still being watched?

Why this matters: Protection budgets are limited. Picking places that complement each other — each adding species the others lack — protects the most breeding species for the fewest sites.

Just 32 grid squares hold recent breeding for 90% of NSW's native species

Squares chosen one at a time to add the most new species (Atlas 2000–2020 + DES 2021–2025)





ANSWER

- 372 native species have recent breeding evidence, spread over 1,566 grid squares.
- Just 3 squares hold breeding for half of these species, and 32 squares hold 90%. The first three chosen are near Gosford (122 species), near Moree (103 species), near Kempsey (116 species).
- 39 species have recent breeding evidence from only one square, so a single local loss could remove them as breeders from the records (listed in the accompanying table).
- Only 22 of the 32 priority squares had any survey in 2021–2025; 10 have gone unwatched in recent years.

What it means: The 32 priority squares are a short, practical list for protection checks and for repeat surveys. The unwatched priority squares should be the first places sent new volunteers.

Caution: Squares rich in species are also often the most surveyed, so some may rank highly partly because they were watched more closely. Species recorded breeding in only one square may simply be under-surveyed.

SUMMARY

What this means for protecting bird species

Noisy Miners	Where miners are present, small woodland birds are recorded 7% less often and their odds of breeding are 10% lower; large birds are unaffected.
Introduced birds	The Common Myna went from 10% to 24% of squares and is moving inland; sparrows, starlings and goldfinches retreated.
Winners and losers	9 of the 12 native birds that lost most ground eat insects, such as Black-faced Woodswallow and Hooded Robin; none of the 12 biggest winners do.
Migrants	79% of summer migrants did worse than typical, against 44% of non-insect-eating residents; insect-eaters that stay all year also slipped.
Where to protect	32 grid squares hold recent breeding for 90% of native species; 10 of them had no survey in 2021–2025.

SUGGESTED NEXT STEPS

1. Trial miner management and shrub-layer planting in woodland remnants, with before-and-after nest surveys.
2. Focus Common Myna trapping on the inland edge of its range, before it settles in new towns.
3. Brief the managers of open woodland and native grassland on the insect-eating birds losing ground.
4. Share the migrant results with national and international migratory-bird programmes.
5. Send volunteers first to the 10 unwatched priority squares, and check the single-square breeders on the ground.

ABOUT THE DATA AND METHOD

- Records: 6,619,049 usable species records in NSW from the Bird Atlas and the DES backlog. eBird is excluded, as agreed with the client.
- Q1–Q4 compare the Atlas with itself, 1977–1999 vs 2000–2020, using full-list surveys (at least 10 species). Atlas and DES rates are never mixed.
- Recent survey lists are shorter and early sheets were more often kept open for months. Q2–Q4 therefore draw 5 single-month lists per grid square per period, average 25 random draws, and compare each species with the typical species.
- Q1 compares surveys with and without Noisy Miners in the same grid square, period and list length (a Mantel–Haenszel matched comparison), with large birds as a control group.
- Q5 uses any recent breeding record (surveys and collections) and a step-by-step complementarity selection.
- All settings can be changed in the accompanying notebook; tables for every chart are in the outputs folder.